# CHARLES MAXIMILLIAN THORN

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## EDUCATION

**Stevens Institute of Technology** | Hoboken, NJ — Expected May 2029
*Bachelor of Engineering, Mechanical Engineering | Minor in Aerospace Engineering*
**GPA:** 3.5
**Coursework:** Engineering Statics (ENGR 211), Thermodynamics (ME 234), Engineering Design I & II (ENGR 111, 122), Circuits & Systems (ENGR 245), Differential Equations, Linear Algebra & Vectors, Multivariable Calculus I

## SKILLS

**Tools:** SolidWorks (proficient), FDM 3D Printing (design, slicing, fabrication), OpenRocket, Claude Code (proficient)
**Programming:** Arduino IDE (proficient), C++ (basic)
**Engineering:** Design for Manufacturing, Iterative Prototyping, FMEA, Risk Assessment

## ENGINEERING EXPERIENCE

**NASA University Student Launch Initiative** | Hoboken, NJ — Sep. 2026 – Present
*Safety Officer (Cross-Team), Vehicle Team (Recovery Subgroup) | OpenRocket*

- Ran FMEA and risk assessments as 1 of 2 safety officers across all subteams and led the Personnel Hazard Analysis, citing OSHA and labor law, to enforce NAR/TRA safety code and NASA USLI requirements.
- Lead R&D on ejection charges for the two-stage dual-deployment recovery system on a two-person subteam: calculated charges per stage, barometer-triggered events, and 20% safety factor backups firing 1 s after the main.
- Support recovery testing of shock cords, drogue and main parachutes, and model decouplers and bulkheads in SolidWorks for the final launch vehicle assembly.

**SAE Aero Design Club** | Hoboken, NJ — May 2026 – Present
*Structures Subteam Member, Micro Class | SolidWorks*

- Design the aerodynamically critical forward fuselage section in SolidWorks, optimizing structural strength, weight, and drag.
- Engineer the longeron and bulkhead layout for laser-cut manufacturability, minimizing construction complexity.

**Stevens Institute of Technology, MOCORE Lab** | Hoboken, NJ — Jun. 2026 – Sep. 2026
*Undergraduate Student Researcher (Raviraj Nataraj, PhD)*

- Took initiative to redesign a lacking aspect of an existing PhD VR rehabilitation setup, owning scope, budget, and timeline.
- Engineered a dual inflatable TPU bladder and 3D-printed trough housing for the forearm jig via iterative prototyping.

## PROJECTS

**Robot Arena** | ENGR 122 — Jan. 2026 – May 2026
*SolidWorks, Arduino, FDM 3D Printing*

- Designed a unique main housing frame in SolidWorks, exceeding spec for minimum weight and maximum strength, printing all parts in 2 hr 10 min against a 4 hr limit.
- Ranked first in preliminary standings and advanced to the finals of the Gallois Autonomous Robot Competition, 1 of 12 teams from roughly 650 first-year engineers.
- Helped write and optimize checkpoint pathfinding using ultrasonic feedback and ArUco marker localization.

**Automatic Plant Watering System** | ENGR 111 — Sep. 2025 – Dec. 2025
*Arduino IDE, SolidWorks*

- Led SolidWorks and Arduino development of an automated soil moisture monitor with wireless reporting to Stevens; iterated the enclosure for print time and strength.

## WORK EXPERIENCE

**Blue Line Pool Management** | Ridgefield, NJ — Jul. 2024 – Present
*Lifeguard, Exclusive Pool Management*

- Supervise independent pool operations at luxury residential properties, enforcing Red Cross CPR, AED, and safety compliance.

## LEADERSHIP & EXTRACURRICULARS

**American Institute of Aeronautics and Astronautics (AIAA)** | Hoboken, NJ — Sep. 2025 – Present
*Secretary (AY 2026–2027)*

- Drive chapter operations: distribute minutes, produce presentations, and build scheduling forms.

**Stevens Institute of Technology, Track & Field** | Hoboken, NJ — Sep. 2025 – Present
*NCAA Div III Student-Athlete, Cross Country & Track (Winter, Spring)*
**Memberships:** AIAA, ASME, National Society of Black Engineers (NSBE), Stevens Commuter Student Union